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# **Analytics Assignment 3: Cluster Analysis**

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# 1 Exploratory Data Analysis

The data set for this assignment includes several variables for listings that appear on the Airbnb site for the City of Cape Town. Table 7 in Appendix A contains a full list of variables, their data types, and descriptions.

## 1.1 Data Cleaning

The Airbnb listings data set contains 16 317 observations across 40 variables, with no duplicate listings identified. Several variables were removed prior to analysis. The first three variables, relating to host identification and naming, were excluded as host-level information is not relevant to listing-level clustering. Similarly, the total host listings count was removed as it characterises the host rather than the listing itself. Of the two neighbourhood variables, `Neighbourhood_Proper & Trim` and `Neighbourhood_Cleansed`, only one was retained as they capture the same information redundantly.

Listings with no revenue generated over any period were removed, as these observations produced zeroed values across most variables and would distort cluster formation. The room type variable was transformed into four binary dummy variables representing entire home, private room, shared room, and hotel room. An occupancy rate variable was constructed to measure listing popularity, after which the individual `Booked` and `Availability` variables were removed. Review scores across the individual categories were averaged into a single composite rating variable. After removing observations with missing values, the final data set contains 14 806 listings across 21 variables.

## 1.2 Data Visualisation

To examine how variables in the dataset relate to one another, several visualisation methods are employed. Figure 1 shows the distribution of daily prices. A strong right-skew is observed, with the majority of listings priced below R10,000 per night. This suggests the presence of outliers, which may influence clustering methods and should therefore be considered in subsequent analysis.

In Figure 1, the neighbourhood variable is used to examine the relationship between daily rate and selected explanatory variables, including the number of bedrooms, bathrooms, and guest capacity. A clear positive relationship is observed between accommodation capacity and daily rate. This is expected, as larger properties typically offer more amenities and space, justifying higher prices. Most listings have relatively high average review scores, with only a few listings recording an average rating below 3.5. Interestingly, there does not appear to be a strong linear relationship between review score and price, suggesting that guest satisfaction is not strongly reflected in pricing.

There is also evidence of a non-linear relationship between the number of annual bookings and price. Prices initially increase as listings become more popular, but tend to decline once

availability becomes constrained. This may reflect strategic pricing behaviour by hosts: less popular listings may reduce prices to attract demand, while highly demanded properties may increase prices. Conversely, some affordable listings remain consistently booked, indicating high demand despite lower pricing.

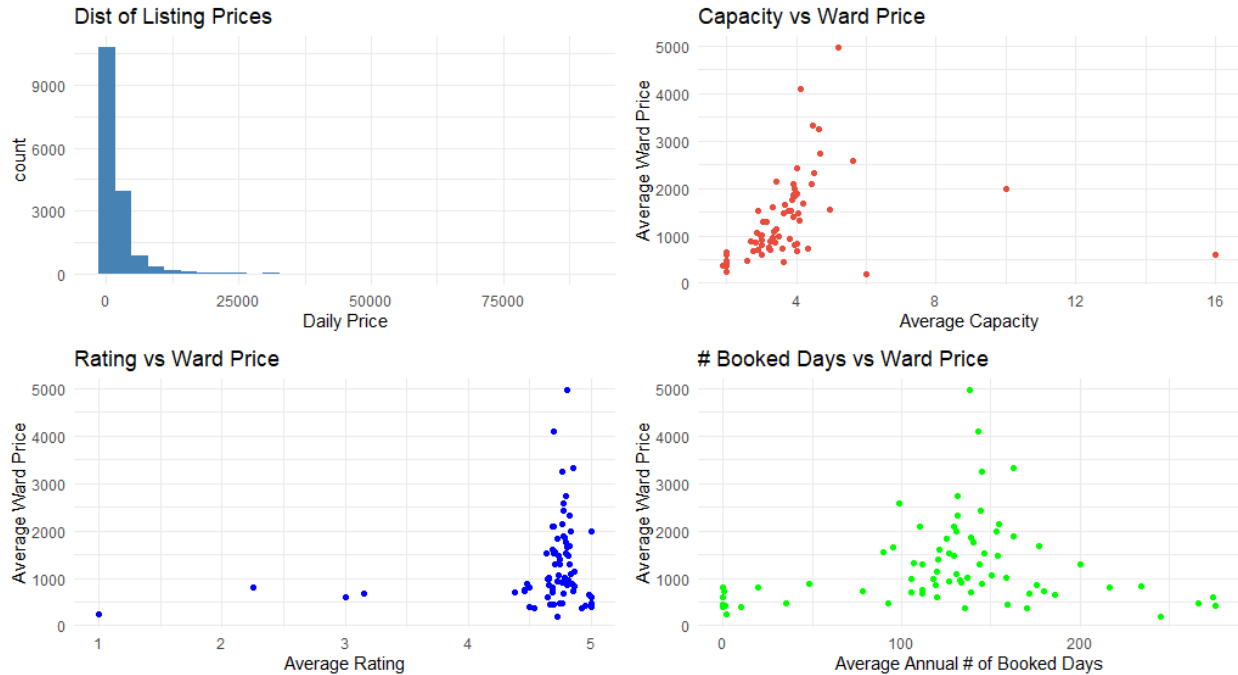


Figure 1: Variable-Variable plots for the average prices of Airbnb listings

Figure 2 visualises the average daily rates across Airbnb neighbourhoods in the City of Cape Town. The most expensive neighbourhood is Rondebosch, followed by Sea Point/Green Point, Camps Bay, and Constantia. Other neighbourhoods exhibit broadly similar price levels, which may make them more difficult to distinguish in clustering analyses.

### 1.3 Principle Component Analysis

Considering the large number of variables to consider, we may implement a Principle Component Analysis as a dimension reduction technique. By using the Elbow method in Figure 3, we may be advised to use the first 5 principle components. The first principle component had the highest percentage of variance explained, with high loadings on the **price**, **Revenue\_365**, **Bedrooms**, **Bathrooms** and **Accommodates**. It is clear that this PC measures the scale of a listing, separating high value properties from cheaper ones, considering both their size and revenue. The next PC had high loadings on the number of reviews, measuring booking frequency, regardless of listing price. PC3 measures a listing's short-term yield, with high loadings on the 30, 60 and 90 day revenue variables. The fourth PC measures a guest's experience by loading high on the review rating variables. The fifth and sixth principle components act as a trigger for property type and the listing's location respectively.

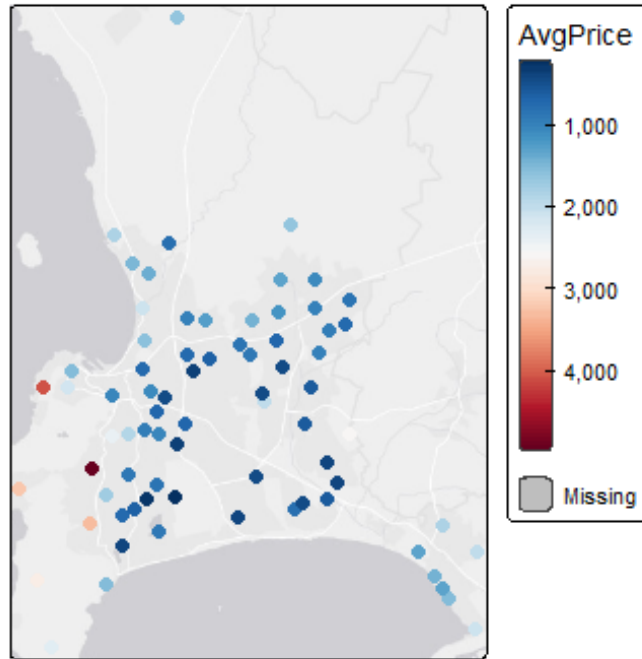


Figure 2: A visualisation of the Airbnb neighbourhoods of the City of Cape Town with their respective average daily rate

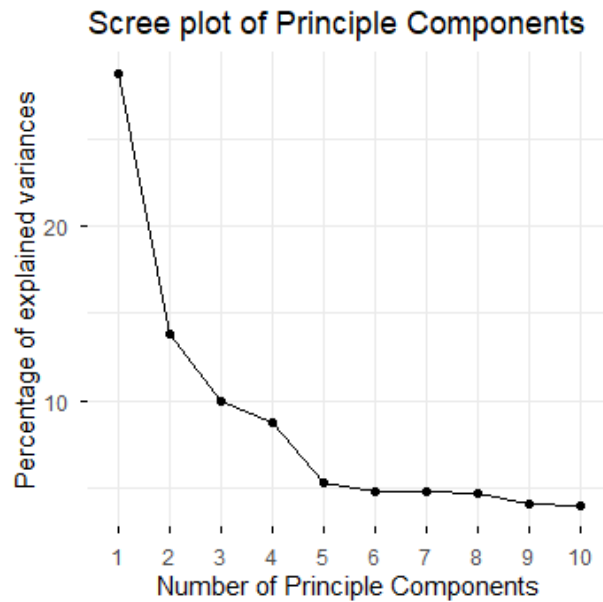


Figure 3: A scree plot visualising the percentage of variance explained from the principle components

Considering this, the following variables are included in the final data set to be used for the analysis: Price, Bedrooms, Bathrooms, Reviews\_per\_Month, avg\_score, occupancy\_score,

pvt\_room, shared\_room, and hotel\_room

These variables are scaled to have comparable ranges before clustering.

## 2 Hierarchical Clustering

To select an appropriate hierarchical clustering method, we compared the stability of the suggested number of clusters across single linkage, complete linkage, and centroid linkage. Complete linkage gave the most stable and interpretable results, so it was used for the final hierarchical clustering analysis.

### 2.1 Limitations with our Hierarchical Clustering

The number of clusters,  $k$ , was assessed using the elbow method, average silhouette method, and Gap statistic. Because hierarchical clustering is computationally expensive for a large data set, a stratified sample was used for the more intensive methods, including the average silhouette method and the Gap statistic. These limitations should be considered when interpreting the final clusters, although the diagnostic results were broadly consistent.

### 2.2 Analysis for the Number of Clusters Chosen

To choose the number of clusters,  $k$ , we compared the elbow method, average silhouette method, and Gap statistic. The elbow method evaluates how the within-cluster sum of squares changes as the number of clusters increases from 2 to 10. The results are shown in Figure 4.

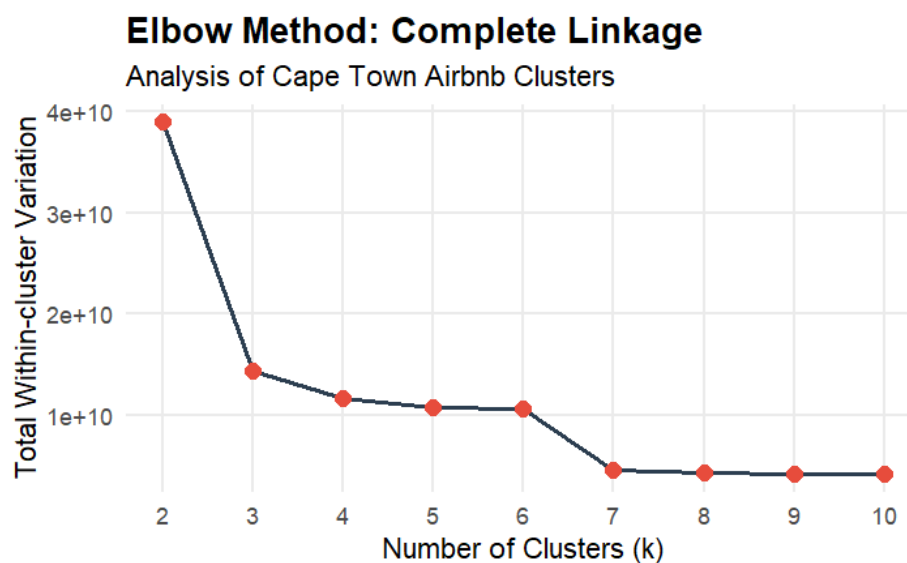


Figure 4: Elbow method for selecting the number of clusters in the complete linkage model

Figure 4 shows a sharp drop in the total within-cluster variation from the 2nd to the 3rd cluster. There also seems to be a drop from the 3rd to the 4th cluster and a noticeable step down from the 6th to the 7th cluster. We therefore chose our value for  $k$  to be 3. Even though there are subsequent drops later in the plot, the most significant elbow occurs at the 3rd cluster, after which the variation begins to stabilize. We did not choose 7 clusters as that appeared to be overfitting the dataset.

The average silhouette method was then used as a second diagnostic for choosing the number of clusters.

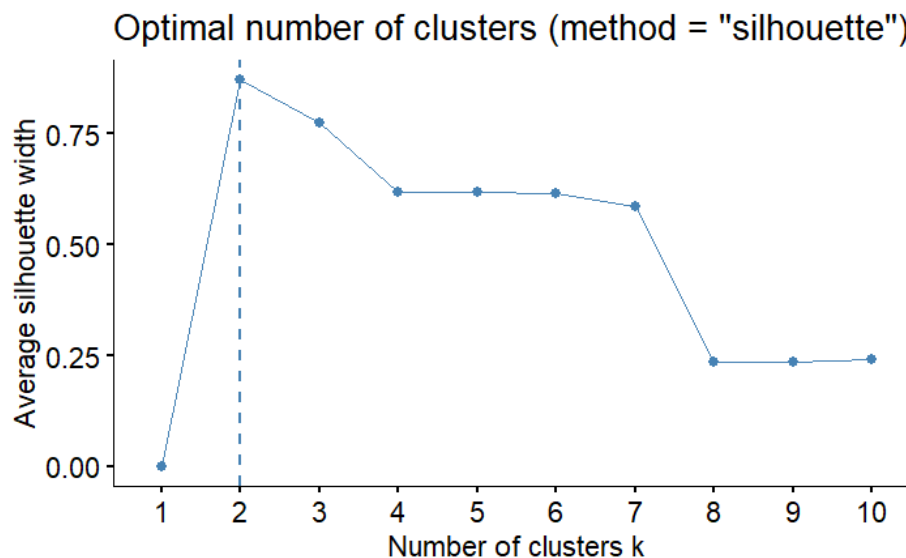


Figure 5: Average silhouette method for selecting the number of clusters in the complete linkage model

Figure 5 shows that the highest average silhouette width occurs at 2 clusters, indicating the strongest mathematical separation. However, a 2-cluster solution is too broad for identifying useful Airbnb market segments. The silhouette width remains high at 3 clusters before declining for larger values of  $k$ , so 3 clusters provides a better balance between cluster quality and interpretability.

The last method we used was the gap statistic to determine the number of clusters to use.



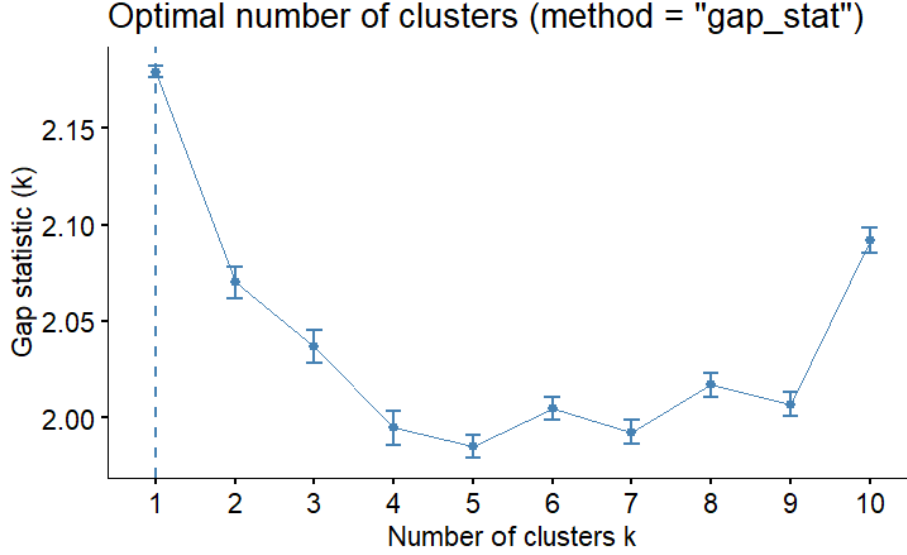


Figure 6: Gap statistic for selecting the number of clusters in the complete linkage model

Figure 6 shows that the optimal value for the number of clusters should be 1, which is not meaningful in determining the listing segments for our Airbnb dataset. There seems to be large changes between the gap statistics for the number of clusters being 1 to 3, after which the downward trend begins to slow down as it approaches 4 and 5 clusters. After 5 clusters, the gap statistic fluctuates and is not presenting stable results, which may indicate overfitting and noise. We therefore decided to pick 3 clusters, as this represents the point right before the metric levels out and becomes unstable at higher cluster counts.

Based on the results above, we decided that our  $k$  value should be 3 as this was the most consistent cluster value across all our diagnostic methods.

## 2.3 Metrics for clusters for Complete Linkage Method

The complete linkage dendrogram was therefore cut to form 3 clusters. The cophenetic correlation was 0.81, indicating that the complete linkage structure preserved the original pairwise distances reasonably well. The resulting clusters were plotted in two dimensions for visual inspection.

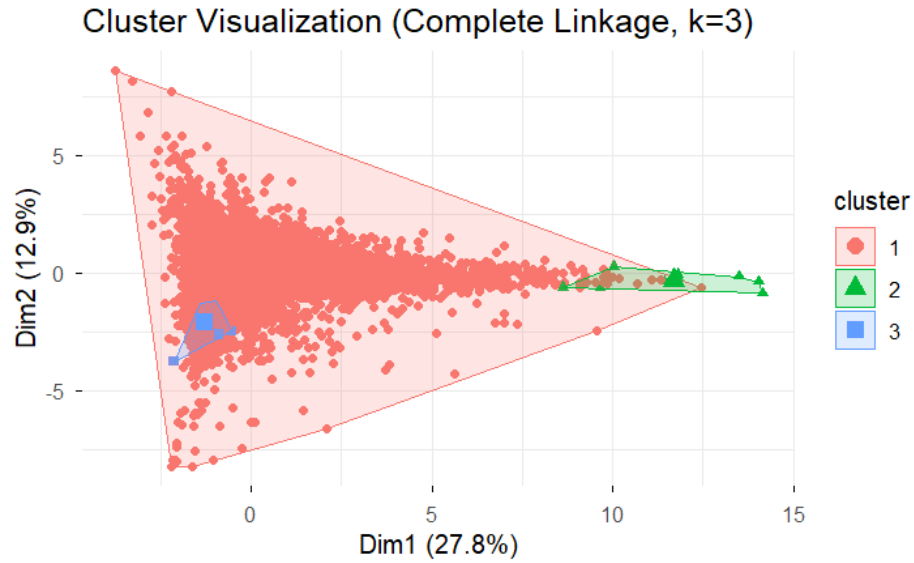


Figure 7: Cluster plot for the complete linkage hierarchical clustering solution

The two plotted dimensions explain less than 50% of the variation in the data, so the plot should be interpreted as a visual summary rather than a complete representation of the clustering structure. Figure 7 shows that clusters 2 and 3 are very small, while cluster 1 contains almost all listings. This suggests that the complete linkage solution mainly separates a few unusual listings rather than producing balanced market segments.

A silhouette analysis was also conducted on the full set of clustering observations and selected variables. The results are shown below.

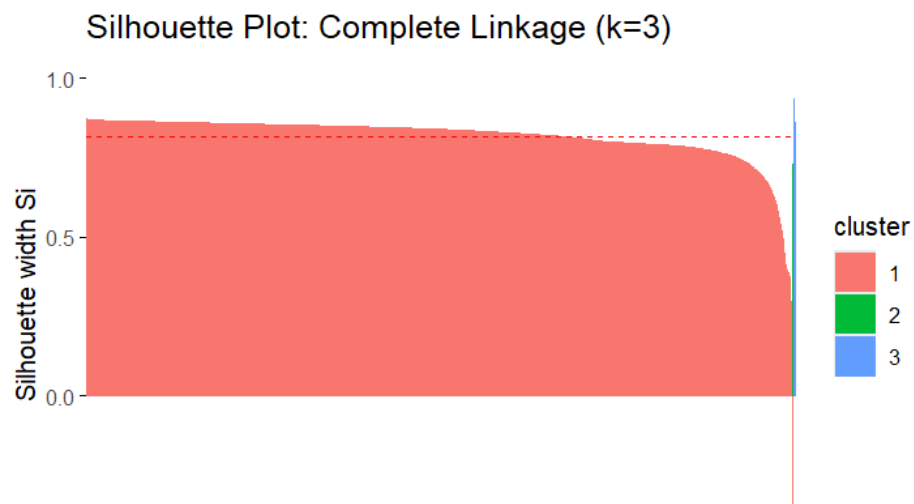


Figure 8: Silhouette plot for the complete linkage clusters on the full data set

Table 1: Cluster sizes and silhouette summary for the complete linkage solution

|                          | Cluster 1 | Cluster 2 | Cluster 3 |
|--------------------------|-----------|-----------|-----------|
| Cluster size             | 14 761    | 10        | 23        |
| Average silhouette width | 0.816     | 0.672     | 0.918     |

---

|                             | Min.   | 1st Qu. | Median | Mean  | 3rd Qu. | Max.  |
|-----------------------------|--------|---------|--------|-------|---------|-------|
| Individual silhouette width | -0.360 | 0.799   | 0.840  | 0.816 | 0.856   | 0.938 |

Figure 8 and Table 1 show generally high average silhouette widths, especially for the two small clusters. However, the negative minimum silhouette width indicates that some observations in the large cluster may be poorly assigned.

Although the silhouette widths are relatively high, the cluster sizes are highly unbalanced: cluster 1 contains 14 761 listings, while clusters 2 and 3 contain only 10 and 23 listings respectively. This suggests that the smaller groups are likely outlier clusters, and that complete linkage does not provide a useful segmentation of the Airbnb listings for this data set.

## 2.4 Cluster profiling: Linkage method

The average profiles for each cluster using the complete linkage method were calculated and are shown below:

Table 2: Average cluster profiles for the complete linkage hierarchical clustering solution

| Cluster | Price     | Bedrooms | Bathrooms | Reviews_per_Month |
|---------|-----------|----------|-----------|-------------------|
| 1       | 2 236.70  | 1.82     | 1.69      | 0.96              |
| 2       | 64 485.00 | 4.00     | 4.10      | 0.29              |
| 3       | 410.22    | 1.00     | 1.26      | 0.46              |

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| Cluster | avg_score | occupancy_score | shared_room | pvt_room | hotel_room |
|---------|-----------|-----------------|-------------|----------|------------|
| 1       | 4.79      | 39.02           | 0.00        | 0.11     | 0.01       |
| 2       | 4.90      | 56.36           | 0.00        | 0.00     | 0.00       |
| 3       | 4.72      | 29.00           | 1.00        | 0.00     | 0.00       |

Table 2 shows that Cluster 2 contains luxury high-end listings with an average of 4 bedrooms and 4.1 bathrooms. This cluster also has the highest occupancy score of the three on average and the highest average score. This makes sense, as high-end listings would likely offer more premium customer service in comparison. Although this is the case, Cluster 2 only has an average of 0.29 reviews per month.

Cluster 1 seems to represent apartments, with a small percentage of listings being private rooms or hotel rooms. The prices are high, but not as expensive as the second cluster.

Cluster 1 offers around 2 bedrooms and 1.7 bathrooms on average, with the highest review rate per month. This cluster also has the second-highest occupancy score.

Cluster 3 seems to offer lower-end listings, with the majority being shared room listings with 1 bed and 1.26 bathrooms. It has the lowest occupancy score and the lowest average score.

### 3 Non-Hierarchical Clustering

For the non-hierarchical clustering analysis, we compared the following algorithms:

- K-means
- K-means++
- PAM
- DBSCAN
- CLARA

The K-means, K-means++, and PAM algorithms mainly produced 2-cluster solutions. Although these clusters were mathematically distinct, they were too broad to provide detailed insight into different listing segments. CLARA was also evaluated, but it did not produce consistent or meaningful clusters for this analysis.

We therefore decided to use DBSCAN for our final non-hierarchical clustering and analysis. DBSCAN was chosen because it does not require us to pre-specify the number of clusters and instead groups data based on density. A major advantage of DBSCAN for this dataset is its ability to automatically identify and isolate outliers as noise without letting them affect the core clustering segments. For the other mentioned algorithms, these extreme outliers would be forced into clusters, which would pull the cluster centers away from the rest of the data and severely distort our profiling and listing segmentation.

#### 3.1 DBSCAN usage and results

The key DBSCAN parameters are `eps` and `MinPts`. `MinPts` was set to 5, which is within the commonly used range of 5 to 10 and matches the dimensionality used in the clustering analysis. The `eps` parameter, which defines the radius of the neighbourhood around each observation, was selected using the elbow method on a k-nearest-neighbour distance plot. The dotted line in Figure 9 indicates the lower-bound reference used when assessing possible `eps` values.

### DBSCAN Parameter Selection: k-Distance Plot

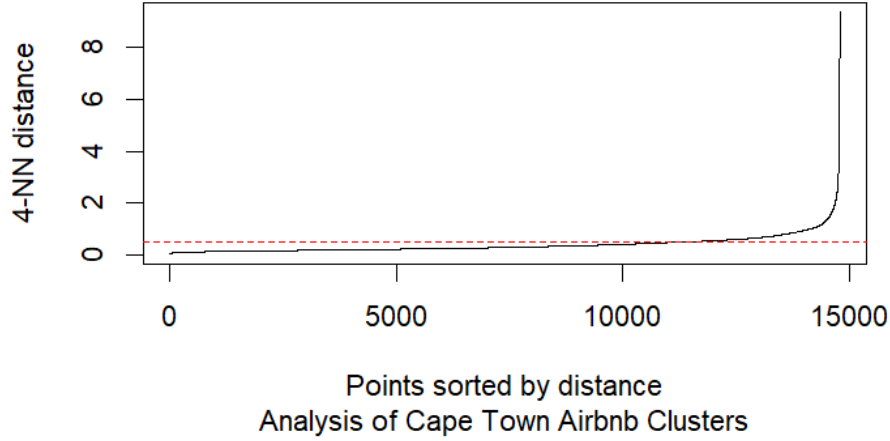


Figure 9: DBSCAN clustering plot for the Airbnb listings data

Figure 9 shows that the 4-nearest-neighbour distance increases gradually before rising sharply near the end of the ordered observations, forming an elbow. The plausible elbow range appears to lie between approximately 0.5 and 3, so DBSCAN was tested using `eps` values of 1.0, 1.5, 2.0, 2.5, and 3.0. The value `eps = 3.0` was selected because it produced the lowest number of outliers while avoiding negative silhouette widths. The results are summarised below.

Table 3: Cluster sizes and silhouette summary for the DBSCAN solution

|                          | Cluster 1 | Cluster 2 | Cluster 3 | Cluster 4 |
|--------------------------|-----------|-----------|-----------|-----------|
| Cluster size             | 12 996    | 1 679     | 73        | 23        |
| Average silhouette width | 0.324     | 0.491     | 0.866     | 0.918     |

|                             | Min.  | 1st Qu. | Median | Mean  | 3rd Qu. | Max.  |
|-----------------------------|-------|---------|--------|-------|---------|-------|
| Individual silhouette width | 0.003 | 0.298   | 0.344  | 0.347 | 0.382   | 0.937 |

Table 3 shows that cluster 1 contains the largest number of listings, while cluster 2 forms a smaller but still substantial segment. Clusters 3 and 4 are much smaller, but both have strong average silhouette widths, suggesting that they are well separated from the other observations. Although clusters 1 and 2 have lower average silhouette widths than clusters 3 and 4, the individual silhouette summary remains positive, indicating that the DBSCAN solution avoids clearly misassigned clustered observations. The remaining observations identified as noise are treated as outliers by DBSCAN and are shown separately in the cluster plot.

The plot below represents the DBSCAN clusters as well as the outliers in the data set.

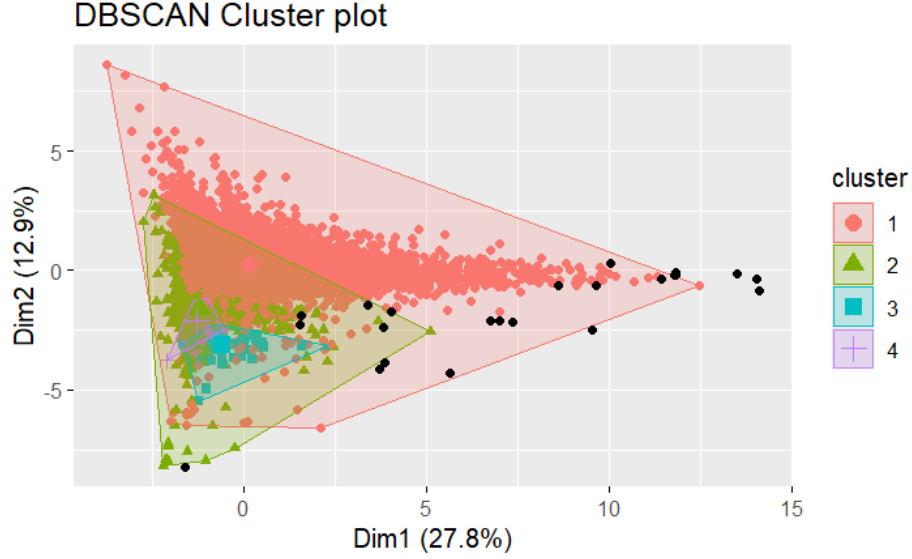


Figure 10: DBSCAN cluster plot showing listing clusters and outliers

The two plotted dimensions explain around 40.7% of the variation in the data, so the figure should be interpreted as a visual summary. Compared with the complete linkage solution, DBSCAN produces more meaningful segmentation because it identifies a second substantial cluster and separates noise points rather than forcing all listings into clusters. The black points represent outliers, which are dispersed rather than forming one concentrated region. This supports the use of DBSCAN for this data set, as it can isolate unusual listings while preserving more interpretable core listing segments.

### 3.2 Cluster profiling: DBSCAN

The average profiles were calculated for each cluster and are shown in the table below.

Table 4: Average cluster profiles for the DBSCAN solution

| Cluster | Price    | Bedrooms | Bathrooms | Reviews_per_Month |
|---------|----------|----------|-----------|-------------------|
| 1       | 2 402.44 | 1.91     | 1.75      | 1.00              |
| 2       | 909.16   | 1.14     | 1.20      | 0.64              |
| 3       | 1 708.92 | 1.37     | 1.21      | 0.24              |
| 4       | 410.22   | 1.00     | 1.26      | 0.46              |

| Cluster | avg_score | occupancy_score | shared_room | pvt_room | hotel_room |
|---------|-----------|-----------------|-------------|----------|------------|
| 1       | 4.80      | 39.52           | 0.00        | 0.00     | 0.00       |
| 2       | 4.78      | 35.35           | 0.00        | 1.00     | 0.00       |
| 3       | 4.76      | 32.52           | 0.00        | 0.00     | 1.00       |
| 4       | 4.72      | 29.00           | 1.00        | 0.00     | 0.00       |

The table above shows that Cluster 1 represents high-end listings of entire homes or apart-

ments. These listings on average have 1.91 bedrooms and 1.75 bathrooms, with exactly 1.00 review per month on average. This cluster features the highest average score (4.80) and the highest occupancy rate (39.52).

Cluster 2 represents mid-range listings, with the majority of them being private room listings offering 1.14 bedrooms and 1.20 bathrooms. This cluster of listings experiences an average of 0.64 reviews per month and maintains a relatively high average score and occupancy score .

Cluster 3 represents higher-priced specialized listings, where the majority of the properties are categorized as hotel rooms. On average, these feature 1.37 bedrooms and 1.21 bathrooms. While the price point is relatively high , Cluster 3 has the lowest number of reviews per month of all four clusters.

Lastly, Cluster 4 represents budget listings that are entirely shared room listings, offering 1 bedroom and 1.26 bathrooms. This cluster is highly similar to the budget cluster identified by the complete linkage algorithm, capturing the lowest occupancy score and the lowest average score.

## 4 Self-Organising Maps

### 4.1 Model Quality

Model quality was assessed using training plots, node counts, the U-matrix, and quantization error. As the model iterates, a node's weights becomes similar to the samples contained within it. So we may expect the distance between node weights and node samples to decrease as the iterations increase until a plateau is reached. Figure 11 illustrates a convergence of this distance after 10,000 iterations.

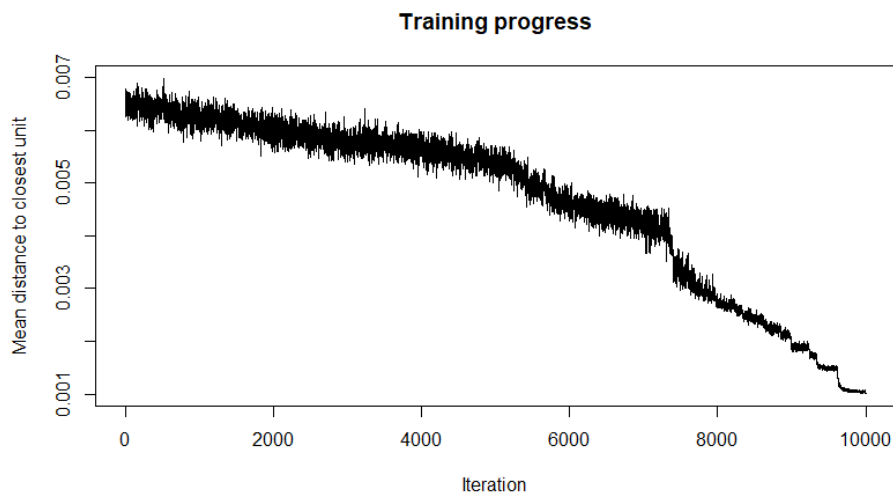


Figure 11: Training progress against the error (difference between samples and node weights)

From the counts plot in Figure ?, the observations appear to be reasonably well distributed across the SOM grid, with each node containing approximately 8 observations on average. This suggests that the map is being utilized efficiently, without excessive concentration in only a few nodes.

To evaluate the similarity between neighbouring nodes, the U-matrix was examined. Figure ? shows that most nodes exhibit low distances to their neighbours, indicating that neighbouring nodes possess highly similar profiles and that the SOM has converged effectively. A small number of nodes display noticeably larger distances from surrounding nodes, suggesting the presence of naturally distinct groups of listings.

Quantization error was used to assess how well the codebook vectors represent the assigned observations. This metric measures the average distance between each observation and its corresponding Best Matching Unit (BMU). Figure ? displays the distribution of quantization errors across the map. The mean quantization error across observations was approximately 0.3, with a maximum value of 5.3. Overall, this indicates that the SOM provides a good representation of the data, although a small number of nodes exhibit comparatively high representation error.

Given the overall quality of the SOM, the resulting node representations appear suitable for subsequent clustering analysis.

## 4.2 Variable Results

To examine the distribution of variables across the SOM, Figure ? presents heatmaps for each variable.

The **Price** variable displays a small number of units with substantially higher pricing than the remainder of the map. This suggests that most listings share relatively similar pricing characteristics, while a limited number of nodes correspond to more expensive listings. A similar pattern is observed for **Reviews\_per\_Month**, where only a few units exhibit notably high values. However, these units are located in different regions of the map from the high-price and large-scale listing nodes, indicating that review activity is not strongly associated with listing size or price.

The **Bedrooms** and **Bathrooms** variables show a dissipation of concentration across the map with larger scale listings in one corner with the size decreasing towards the top left corner.

The **avg\_score** variable shows consistently high ratings across most nodes, with only a small number of units reporting comparatively low review scores. In contrast, the occupancy score demonstrates the least spatial continuity across the map, with high and low values dispersed throughout the SOM rather than concentrated in contiguous regions. This suggests that occupancy behaviour varies across multiple listing profiles and is not driven by a single dominant characteristic. The average score is lower in hotel room and shared listings indicating visitor preferences away from traditional stays.

Finally, the room-type variables indicate strong separation between listing categories. The **shared\_room**, **pvt\_room**, and **hotel\_room** variables each concentrate within distinct regions



of the SOM, demonstrating that the model effectively discriminates between room types. The units outside of these variables' concentration shows entire home listings.

### 4.3 Clustering

To create clusters from the SOM, two clustering methods were considered: K-Means and Hierarchical (Complete Linkage) Clustering. On average, hierarchical clustering yielded significantly lower within-cluster error scores than K-Means, largely because K-Means is sensitive to outliers. Complete linkage hierarchical clustering was therefore selected to partition the SOM nodes into clusters.

To determine the optimal number of clusters, the Elbow method was first considered. Figure 12 displays the total within-cluster sum of squares (WSS) across  $K = 2$  to  $K = 15$ . The plot exhibits an unexpected rise in WSS between  $K = 2$  and  $K = 4$ , followed by a sharp decline at  $K = 5$ , which may reflect instability in the complete linkage algorithm at those values of  $K$  given the SOM node topology. WSS then rises again between  $K = 6$  and  $K = 10$  before dropping sharply at  $K = 11$ , after which the rate of decrease slows considerably. The elbow is therefore most apparent at  $K = 5$  and  $K = 11$ , suggesting that either value may represent a reasonable partition; however, the non-monotonic behaviour of the curve makes it difficult to identify a single unambiguous elbow, and the result should be interpreted with caution..

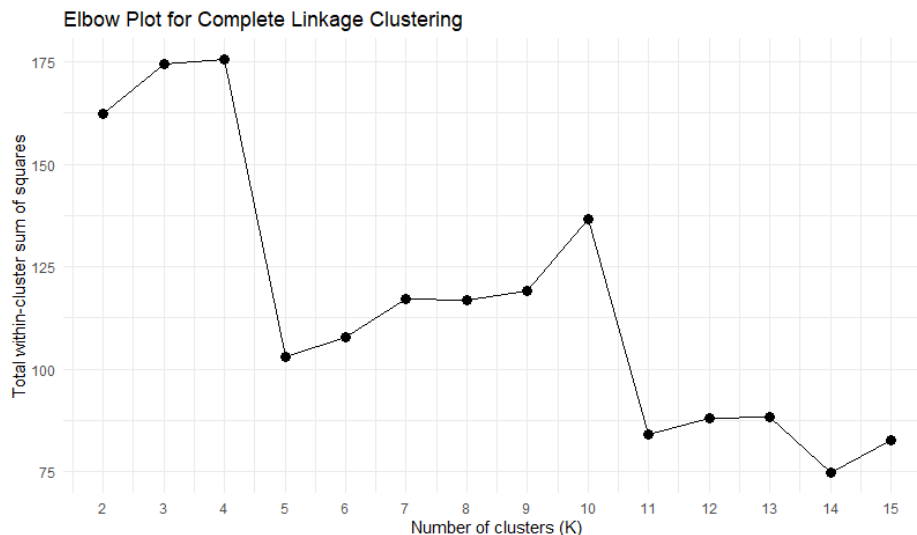


Figure 12: Scree plot of number of clusters formed with their total within-cluster sum of squared residuals

As a second diagnostic, the average silhouette's were produced as displayed in Figure 13.

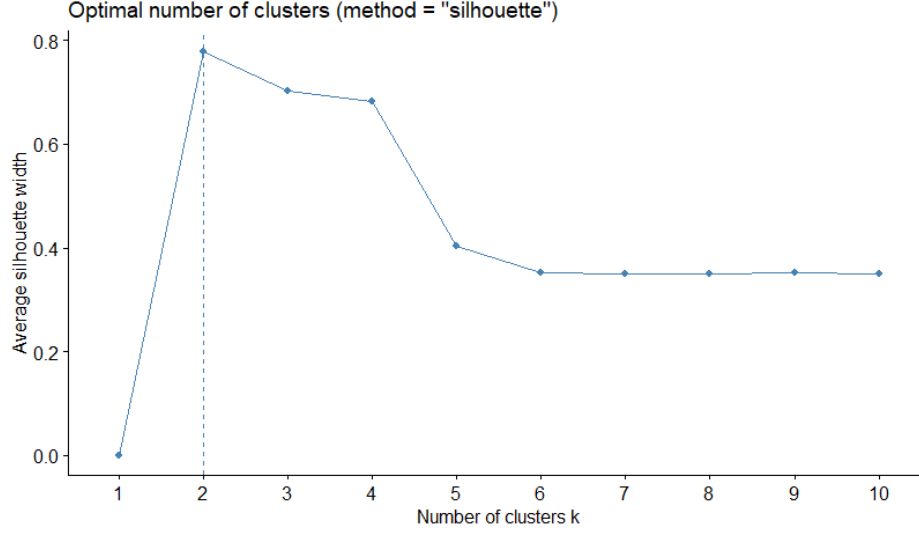


Figure 13: Average silhouette method for selecting the number of clusters in the SOM's complete linkage model

Figure 13 shows that the highest average silhouette width occurs at  $K = 2$  ( $\approx 0.92$ ), suggesting two well-separated clusters. However, a two-cluster solution is too coarse for the purposes of listing segmentation, as it would obscure meaningful distinctions between property types. The silhouette width decreases sharply to  $K = 3$  and  $K = 4$  (both  $\approx 0.79$ ), after which further values of  $K$  yield a more gradual decline. Since  $K = 3$  and  $K = 4$  produce near-identical silhouette scores, the marginal loss in cohesion from choosing  $K = 4$  is negligible, and the additional cluster provides greater discriminatory power between listing segments.

The Gap statistic is considered as the last diagnostic as illustrated below in Figure 14.

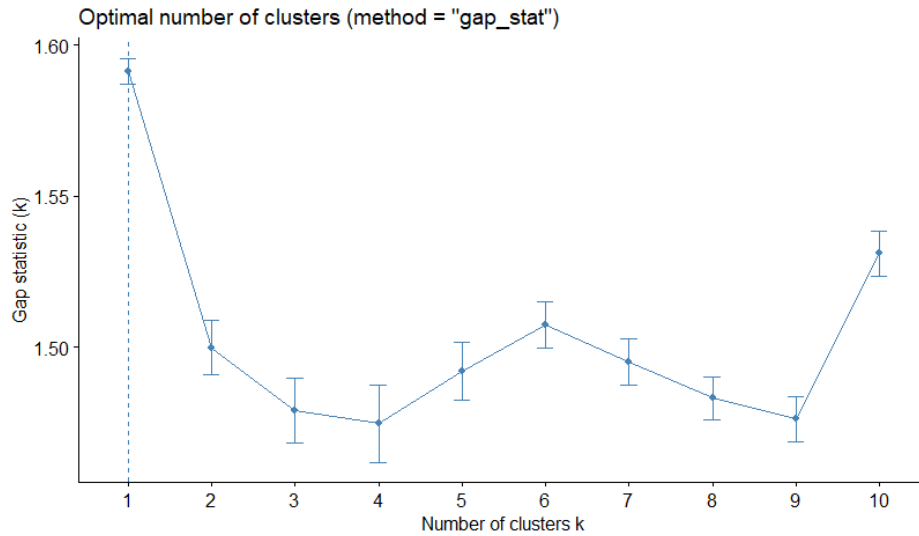


Figure 14: Gap statistic for selecting the number of clusters in the SOM complete linkage model

Figure 14 presents the Gap statistic across  $K = 1$  to  $K = 10$ . The statistic is maximised at  $K = 1$ , with a sharp decline at  $K = 2$  and a continued decrease to a local minimum at  $K = 4$ . A recovery is observed at  $K = 6$ , followed by a further decline to a second local minimum at  $K = 9$ , before rising again at  $K = 10$ . Although the formal Tibshirani criterion nominally selects  $K = 1$ , this is likely an artefact of the SOM pre-processing step: the SOM grid compression reduces the between-cluster variance that the Gap statistic relies upon to discriminate among values of  $K > 1$ . The overlapping standard error bars across  $K = 2$  to  $K = 10$  suggest that no single value of  $K$  is strongly preferred over another in this range. Consequently, the Gap statistic neither confirms nor contradicts the elbow and silhouette results. The final  $K$  is therefore selected based off the two former criteria.

Due to the instability of the complete linkage method, we conclude that a  $k = 4$  is applicable by the average silhouette method. The dendrogram is therefore cut at  $k = 4$  to form four clusters. The clusters are visualised on two dimensions in Figure 15 where the the dimensions capture around 40% of the variation in the listings, so the plot may not be used to make any conclusions. Similar to the 2 previous clustering methods, a large cluster is formed. A difference from the DBSCAN is that the outliers (in the second cluster) have a dedicated cluster which is in agreement with the hierarchical clustering.

Another difference from the DBSCAN is that another cluster is formed rather than a large cluster (within the red area in Figure 10).

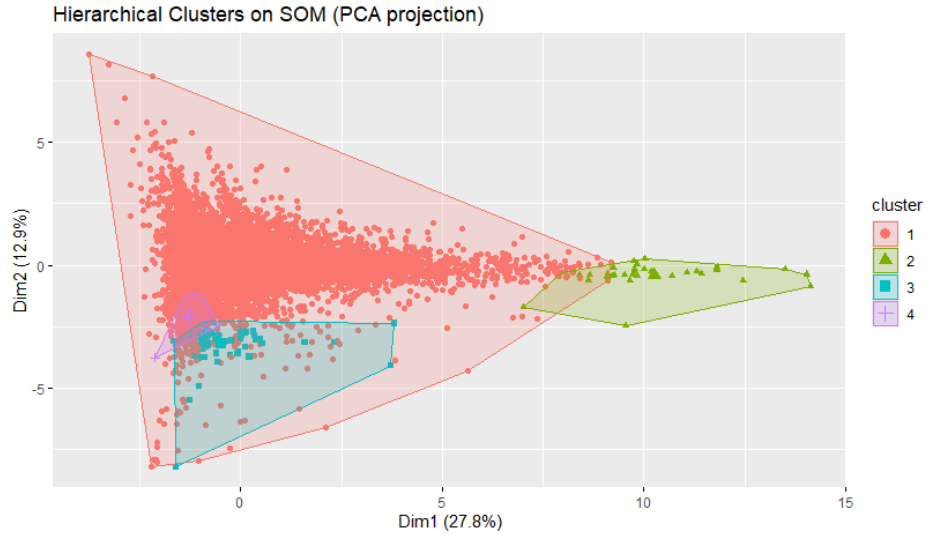


Figure 15: Cluster plot for the complete linkage hierarchical clustering on the SOM

## 4.4 Cluster Profiling

To determine the type of clusters formed and what they represent, cluster profiles are calculated using the variable averages as shown in the table below:

Table 5: Average cluster profiles for the SOM solution

| Cluster | # of Listings | Price     | Bedrooms | Bathrooms | Reviews_per_Month |
|---------|---------------|-----------|----------|-----------|-------------------|
| 1       | 14 662        | 2 193.44  | 1.81     | 1.68      | 0.96              |
| 2       | 33            | 41 080.88 | 5.91     | 6.24      | 0.26              |
| 3       | 76            | 1 905.26  | 1.43     | 1.30      | 0.23              |
| 4       | 23            | 410.22    | 1.00     | 1.26      | 0.46              |

| Cluster | # of Listings | avg_score | occupancy_rate | shared_room | pvt_room | hotel_room |
|---------|---------------|-----------|----------------|-------------|----------|------------|
| 1       | 14 662        | 4.79      | 39.07          | 0.00        | 0.12     | 0.00       |
| 2       | 33            | 4.87      | 33.74          | 0.00        | 0.00     | 0.00       |
| 3       | 76            | 4.71      | 33.23          | 0.00        | 0.00     | 1.00       |
| 4       | 23            | 4.72      | 29.00          | 1.00        | 0.00     | 0.00       |

Cluster 1 represents the typical Airbnb listing profile, characterised by approximately two bedrooms and bathrooms and an average daily rate of  $R2\,193.44$ . This cluster is the largest by far, comprising 14 662 listings (99.1% of the sample), and reports the highest review frequency and occupancy rate amongst all clusters, reflecting the routine demand and turnover associated with standard Airbnb listings.

Consistent with the profiles identified by the preceding methods, Cluster 2 captures a high-end entire homes and apartments segment, comprising only 33 listings with a substantially elevated average daily rate of  $R41\,080.88$ . This cluster also reports the highest average guest rating, which appears to contradict the earlier finding of independence between pricing and review scores. Suggesting that quality may be closely aligned with price at the upper-end.

Clusters 3 and 4 correspond to hotel rooms and shared rooms respectively, each comprising a small number of listings (76 and 23 respectively). Both clusters report comparatively low visitor frequency, with shared rooms recording the lowest occupancy rate and the lowest average daily rate of  $R410.22$  per night, reflecting the budget-oriented nature of this listing type.

## 5 External Validation Metrics: Pairwise Comparisons

Table 6 shows a pairwise comparison of the three methods using the Adjusted Rand Index (ARI) and Jaccard Index. Across all comparisons, the Jaccard Index is consistently high ( $\geq 0.998$ ), reflecting strong agreement in how pairs of observations are co-clustered. However, this is largely driven by the dominant cluster inflating pairwise positive agreements rather than reflecting true structural alignment across all clusters.

Table 6: External validation metrics across all clustering solutions

| <b>Comparison</b> | <b>ARI</b> | <b>Jaccard</b> |
|-------------------|------------|----------------|
| HC vs DBSCAN      | 0.045      | 0.998          |
| SOM-HC vs HC      | 0.397      | 1.000          |
| SOM-HC vs DBSCAN  | 0.108      | 0.998          |

The ARI corrects for chance and penalises disagreement across all cluster boundaries. Using the ARI, the weakest agreement is between HC and DBSCAN ( $\text{ARI} = 0.045$ ), indicating that despite sharing similar pairwise co-clustering, the two methods partition the data quite differently overall due to the differences in distance and density approaches. SOM-HC shows the strongest agreement with direct HC ( $\text{ARI} = 0.397$ ), which is expected given that both methods use complete linkage hierarchical clustering. SOM-HC and DBSCAN show moderate agreement ( $\text{ARI} = 0.108$ ), sitting between the other two comparisons. Overall, the SOM-HC solution is most structurally similar to direct HC, while DBSCAN produces the most distinct partitioning of the three methods.

## 6 Conclusion

The four clusters identified by the SOM solution are broadly consistent with those recovered by both the DBSCAN and hierarchical clustering methods.

The most consistently recovered profile across all three methods is the budget shared room segment. SOM Cluster 4, DBSCAN Cluster 4, and HC Cluster 3 all correspond to shared room listings with a single bedroom, the lowest occupancy rate, and the lowest average daily rate. The near-identical profile across methods shows strong evidence that this segment is represented by a well-separated group in the data.

The hotel room segment is similarly stable across methods. SOM Cluster 3 and DBSCAN Cluster 3 both isolate hotel room listings with comparable bedroom and bathroom counts and relatively low review frequencies. This profile does not emerge as a distinct cluster in the hierarchical solution, where it is likely absorbed into the broader standard listings segment, suggesting that complete linkage clustering is less sensitive to this distinction.

The high-end entire homes segment is recovered by all three methods, though with differing degrees of extremity. Complete linkage Cluster 2 yields the most pronounced high-end profile, with an average daily rate of  $R64\,485.00$ , 4 bedrooms, and 4.10 bathrooms, compared to SOM Cluster 2 ( $R41\,080.88$ , 5.91 bedrooms & 6.24 bathrooms) and DBSCAN Cluster 1 ( $R2\,402.44$ , 4 bedrooms, 4.1 bathrooms). The difference in the HC and the SOM is that the SOM has more observations in the class, and thus represents it more accurately. The less extreme DBSCAN cluster is due to the outliers not being classified into a cluster.

Finally, the standard listings segment (containing most observations) — is captured by SOM Cluster 1 and HC Cluster 1, both characterised by approximately two bedrooms and bathrooms, high review frequency, and moderate-to-high occupancy rates. DBSCAN Cluster 2 partially corresponds to this profile but skews toward private room listings at a mid-range price point, suggesting that DBSCAN draws a finer distinction within what the SOM and hierarchical methods treat as a single dominant cluster.

## 7 Appendix A

Table 7: Variable Descriptions

| Variable          | Description                                                                         |
|-------------------|-------------------------------------------------------------------------------------|
| Price             | <i>Numeric – The nightly rental price of the listing</i>                            |
| Bedrooms          | <i>Numeric – The number of bedrooms in the listing</i>                              |
| Bathrooms         | <i>Numeric – The number of bathrooms in the listing</i>                             |
| Reviews_per_Month | <i>Numeric – The average number of reviews received per month</i>                   |
| avg_score         | <i>Numeric – The average guest review score across the review rating categories</i> |
| occupancy_score   | <i>Numeric – A constructed measure of listing occupancy and popularity</i>          |
| pvt_room          | <i>Binary – Indicates whether the listing is a private room</i>                     |
| shared_room       | <i>Binary – Indicates whether the listing is a shared room</i>                      |
| hotel_room        | <i>Binary – Indicates whether the listing is a hotel room</i>                       |